

E-Commerce Sales Analysis

This dataset consists of orders made in different countries from December 2010 to December 2011. The company is a UK-based online retailer that mainly sells unique all-occasion gifts, with many of its customers being wholesalers.

541909

...	↑↓	I...	...	↑↓	S...	...	↑↓	Description	...	↑↓	...	↑↓	Invoi...	...	↑↓	U	...	↑↓	C...	...	↑↓	Country
1		536365			85123A			WHITE HANGING HEART T-LIGHT HOLDER			6		12/1/10 8:26			2.55			17850			United Kingdom
2		536365			71053			WHITE METAL LANTERN			6		12/1/10 8:26			3.39			17850			United Kingdom
3		536365			84406B			CREAM CUPID HEARTS COAT HANGER			8		12/1/10 8:26			2.75			17850			United Kingdom
4		536365			84029G			KNITTED UNION FLAG HOT WATER BOTTLE			6		12/1/10 8:26			3.39			17850			United Kingdom
5		536365			84029E			RED WOOLLY HOTTIE WHITE HEART.			6		12/1/10 8:26			3.39			17850			United Kingdom
6		536365			22752			SET 7 BABUSHKA NESTING BOXES			2		12/1/10 8:26			7.65			17850			United Kingdom
7		536365			21730			GLASS STAR FROSTED T-LIGHT HOLDER			6		12/1/10 8:26			4.25			17850			United Kingdom
8		536366			22633			HAND WARMER UNION JACK			6		12/1/10 8:28			1.85			17850			United Kingdom
9		536366			22632			HAND WARMER RED POLKA DOT			6		12/1/10 8:28			1.85			17850			United Kingdom
10		536367			84879			ASSORTED COLOUR BIRD ORNAMENT			32		12/1/10 8:34			1.69			13047			United Kingdom
11		536367			22745			POPPY'S PLAYHOUSE BEDROOM			6		12/1/10 8:34			2.1			13047			United Kingdom
12		536367			22748			POPPY'S PLAYHOUSE KITCHEN			6		12/1/10 8:34			2.1			13047			United Kingdom
13		536367			22749			FELTCRAFT PRINCESS CHARLOTTE DOLL			8		12/1/10 8:34			3.75			13047			United Kingdom
14		536367			22310			IVORY KNITTED MUG COSY			6		12/1/10 8:34			1.65			13047			United Kingdom
15		536367			84969			BOX OF 6 ASSORTED COLOUR TEASPOONS			6		12/1/10 8:34			4.25			13047			United Kingdom
16		536367			22623			BOX OF VINTAGE JIGSAW BLOCKS			3		12/1/10 8:34			4.95			13047			United Kingdom

Rows: 100

Expand

Data Dictionary

Variable	Explanation
InvoiceNo	A 6-digit integral number uniquely assigned to each transaction. If this code starts with letter 'c' it indicates a cancellation.
StockCode	A 5-digit integral number uniquely assigned to each distinct product.
Description	Product (item) name
Quantity	The quantities of each product (item) per transaction
InvoiceDate	The day and time when each transaction was generated
UnitPrice	Product price per unit in sterling (pound)
CustomerID	A 5-digit integral number uniquely assigned to each customer
Country	The name of the country where each customer resides

[Source](#) of dataset.

Citation: Daqing Chen, Sai Liang Sain, and Kun Guo, Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining, Journal of Database Marketing and Customer Strategy Management, Vol. 19, No. 3, pp. 197-208, 2012 (Published online before print: 27 August 2012. doi: 10.1057/dbm.2012.17).

Data Exploration

E-Commerce Dataonline_retail.csv

...	↑↓	I...	...	↑↓	Stoc...	...	↑↓	Description	...	↑↓	...	↑↓	Invoic...	...	↑↓	U	...	↑↓	C...	...	↑↓	Cou
10221		537237			15036			ASSORTED COLOURS SILK FAN					2	12/6/10 9:58				1.28	Invalid	Nu...		U
10222		537237			15056N			EDWARDIAN PARASOL NATURAL					1	12/6/10 9:58				12.72	Invalid	Nu...		U
10223		537237			16218			CARTOON PENCIL SHARPENERS					1	12/6/10 9:58				2.13	Invalid	Nu...		U
10224		537237			16238			PARTY TIME PENCIL ERASERS					5	12/6/10 9:58				0.43	Invalid	Nu...		U
10225		537237			16258A			SWIRLY CIRCULAR RUBBERS IN BAG					3	12/6/10 9:58				0.84	Invalid	Nu...		U
10226		537237			17003			BROCADE RING PURSE					2	12/6/10 9:58				0.43	Invalid	Nu...		U
10227		537237			17012C			ORIGAMI LAVENDER INCENSE/CANDL SET					2	12/6/10 9:58				2.51	Invalid	Nu...		U
10228		537237			17012E			ORIGAMI OPIUM INCENSE/CANDLE SET					1	12/6/10 9:58				2.51	Invalid	Nu...		U
10229		537237			17012F			ORIGAMI SANDLEWOOD INCENSE/CAND SET					1	12/6/10 9:58				2.51	Invalid	Nu...		U
10230		537237			18098C			PORCELAIN BUTTERFLY OIL BURNER					1	12/6/10 9:58				5.91	Invalid	Nu...		U
10231		537237			20615			BLUE POLKADOT PASSPORT COVER					1	12/6/10 9:58				4.21	Invalid	Nu...		U
10232		537237			20622			VIPPASSPORT COVER					1	12/6/10 9:58				4.21	Invalid	Nu...		U
10233		537237			20665			RED RETROSPOT PURSE					1	12/6/10 9:58				5.91	Invalid	Nu...		U
10234		537237			20676			RED RETROSPOT BOWL					1	12/6/10 9:58				2.51	Invalid	Nu...		U
10235		537237			20696			FLORAL SOFT CAR TOY					1	12/6/10 9:58				7.62	Invalid	Nu...		U
10236		537237			20699			MOUSEY LONG LEGS SOFT TOY					1	12/6/10 9:58				5.06	Invalid	Nu...		U

Rows: 12,500Truncated from 541,909 rowsExpand

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$ InvoiceNo   : chr [1:541909] "536365" "536365" "536365" "536365" ...
$ StockCode  : chr [1:541909] "85123A" "71053" "84406B" "84029G" ...
$ Description: chr [1:541909] "WHITE HANGING HEART T-LIGHT HOLDER" "WHITE METAL LANTERN" "CREAM CUPID HEARTS COAT HANGER"
"KNITTED UNION FLAG HOT WATER BOTTLE" ...
$ Quantity   : num [1:541909] 6 6 8 6 6 2 6 6 6 32 ...
$ InvoiceDate: chr [1:541909] "12/1/10 8:26" "12/1/10 8:26" "12/1/10 8:26" "12/1/10 8:26" ...
$ UnitPrice  : num [1:541909] 2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85 1.85 1.69 ...
$ CustomerID : num [1:541909] 17850 17850 17850 17850 17850 ...
$ Country    : chr [1:541909] "United Kingdom" "United Kingdom" "United Kingdom" "United Kingdom" ...
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..   StockCode = col_character(),
..   Description = col_character(),
..   Quantity = col_double(),
..   InvoiceDate = col_character(),
..   UnitPrice = col_double(),
..   CustomerID = col_double(),
..   Country = col_character()
.. )
- attr(*, "problems")=<externalptr>
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InvoiceNo

Length:541909

Class :character

Mode :character

StockCode

Length:541909

Class :character

Mode :character

Description

Length:541909

Class :character

Mode :character

Quantity

Min. :-80995.00

1st Qu.: 1.00

Median : 3.00

Mean : 9.55

3rd Qu.: 10.00

Max. : 80995.00

InvoiceDate

Length:541909

Class :character

Mode :character

UnitPrice

Min. :-11062.06

1st Qu.: 1.25

Median : 2.08

Mean : 4.61

3rd Qu.: 4.13

Max. : 38970.00

CustomerID

Min. :12346

1st Qu.:13953

Median :15152

Mean :15288

3rd Qu.:16791

Max. :18287

NA's :135080

Country

Length:541909

Class :character

Mode :character

InvoiceNo: 0 StockCode: 0 Description: 1454 Quantity: 0 InvoiceDate: 0 UnitPrice: 0 CustomerID: 135080 Country:

Data Analysis

Return by Products

Looking at the variable summary, there are negative values in the Quantity and UnitPrice variables. Those in UnitPrice were described as existing bad debts. For negative Quantity values, they could be known as returned/cancelled transactions with positive UnitPrice and InvoiceNo starting with "C". The rest would be error transactions. The one with the highest number of products returned was PAPER CRAFT, LITTLE BIRDIE, with approximately 81,000 items. Ranked second is MEDIUM CERAMIC TOP STORAGE JAR, which has nearly 75,000 items returned (74,215 of which were in one transaction). However, both products were returned in such large quantities in a single transaction by one customer. The returns may have been due to an error or mistake and requires further investigation with additional information, suggesting these are likely data anomalies or bulk order cancellations rather than typical customer return behaviour or repeated customer dissatisfaction.

REGENCY CAKESTAND 3 TIER records the highest number of returns (181), excluding Manual-entry and Postage returns, which have been categorised separately. This indicates a need for further investigation into product quality, customer expectations, or product handling.

...	I...	S...	Description	...	Invoi...	U...	C...	Country
1	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/10 8:26	2.55	17850	United Kin
2	536365	71053	WHITE METAL LANTERN	6	12/1/10 8:26	3.39	17850	United Kin
3	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/10 8:26	2.75	17850	United Kin
4	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/10 8:26	3.39	17850	United Kin
5	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/10 8:26	3.39	17850	United Kin
6	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	12/1/10 8:26	7.65	17850	United Kin

Rows: 6

...	Description	sum...
1	PAPER CRAFT , LITTLE BIRDIE	-80995
2	MEDIUM CERAMIC TOP STORAGE JAR	-74494
3	ROTATING SILVER ANGELS T-LIGHT HLDR	-9376
4	Manual	-4066
5	FAIRY CAKE FLANNEL ASSORTED COLOUR	-3150
6	PINK BLUE FELT CRAFT TRINKET BOX	-2617
7	WHITE HANGING HEART T-LIGHT HOLDER	-2578
8	GIN + TONIC DIET METAL SIGN	-2030
9	HERB MARKER BASIL	-1527
10	FELTCRAFT DOLL MOLLY	-1447

Rows: 10

...	Description	coun...
1	Manual	244
2	REGENCY CAKESTAND 3 TIER	181
3	POSTAGE	126
4	JAM MAKING SET WITH JARS	87
5	Discount	77
6	SET OF 3 CAKE TINS PANTRY DESIGN	74
7	SAMPLES	61
8	STRAWBERRY CERAMIC TRINKET BOX	55
9	ROSES REGENCY TEACUP AND SAUCER	54
10	RECIPE BOX PANTRY YELLOW DESIGN	47

Rows: 10

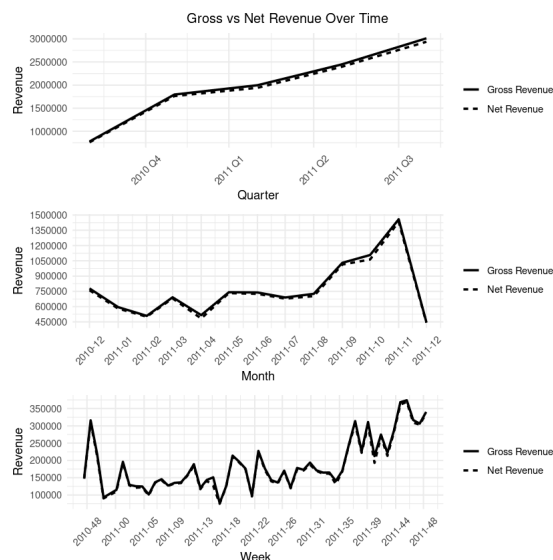
Business Growth and Market Dynamics

The two bulk transactions were considered anomalies; they would be removed to avoid any effect on further analysis. Special transaction codes such as postage, discounts, and manual adjustments were excluded to ensure that the analysis reflected actual product performance and customer behaviour.

Revenue Growth

Revenue shows a growing tendency in the second half of the year, likely driven by seasonal demand due to the holiday season, especially in the last quarter of both years. However, a sharp decline witnessed in December 2011 may indicate a post-peak demand drop. Weekly revenue fluctuations suggest unstable purchasing patterns, possibly driven by bulk orders, promotions, or irregular customer behaviour, particularly during off-peak seasons around the first and second quarters of the year. While the business is growing notably, returns are increasingly offsetting revenue gains due to the widening gap between gross and net revenue in the quarterly view. Return rate is the key driver of reducing net revenue despite growth over time. This indicates the growth does not fully translate to profitability, highlighting the importance of focusing on return management.

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UK vs International Markets

As the business is UK-based, the UK remains the core market while dominating other countries in both volume and revenue, with a moderate average order value (around £425 per order) and a smaller order basket (approximately 218 items per order). Even though international markets contribute relatively lower total orders, they exhibit significantly higher values and much larger basket sizes.

The data shows the UK as a high-volume, low-value market, with more frequent and small-size purchases, while non-UK customers behave quite the opposite, with less frequent, large-volume or wholesale-like purchasing behaviour, possibly due to difficulties that may be encountered when buying/shipping from abroad (time, cost, inventory, etc.). This suggests a need for necessary strategies in terms of pricing, inventory, and logistics management.

Return Behaviour

Some countries show relatively low order volume, which might cause small sample bias and be misleading when interpreting the figures. Therefore, a minimum threshold of 20 has been applied to ensure statistical reliability when calculating return rates.

Return patterns were seen to be strongly market-dependent and vary notably across markets. Italy, Switzerland and Germany are the top three with the highest return rates (40%, 36%, and 30%, respectively). These high-return European markets exhibit potential issues in product fit or customer expectations.

In contrast, the Netherlands stands out with the lowest return rate (~3%), suggesting alignments between supply - product offering and demand - customer expectations. The UK has a moderate return rate (~15%) and stable purchasing behaviour, making it a reliable benchmark for performance evaluations.

Market Segmentation: Value vs Risk

Markets can be segmented based on profitability potential and operational risk, as explicitly witnessed by the market performance. The Netherlands emerges as a high-value, low-risk market, characterised by the highest average order value, the largest basket size, and the lowest return rates. These factors make the Netherlands a high-performing market and an ideal profile for profitability and growing potential.

Conversely, when combining similar factors, Sweden, Switzerland and EIRE show higher risk profiles with larger order sizes, higher order values, and much higher return rates, amplifying the adverse financial impact on the business.

...	↑↓	Country	...	↑↓	total_pu...	...	↑↓	total...	...	↑↓	ret...	...	↑↓	total_r...	...	↑↓	avg_ord...	...	↑↓	...	⋮
15		Netherlands			94			3			3.19			283479.54			3020.1			2136.53	
7		Australia			56			11			19.64			136990			2467.34			1503.71	
18		Japan			19			6			Invalid Numb...			35419.79			1969.28			1369.26	
27		Singapore			4			0			Invalid Numb...			9120.39			2280.1			1308.5	
5		Sweden			34			8			23.53			35176.91			1083.51			1060.06	
26		Hong Kong			7			1			Invalid Numb...			9942.04			1421.86			681.14	
2		Switzerland			50			18			36			52505.35			1061.76			610.56	
14		Norway			32			2			6.25			32292.96			1014.21			602.44	
32		Canada			5			0			Invalid Numb...			3115.44			623.09			552.4	
23		Israel			8			1			Invalid Numb...			7907.82			1016.91			551.12	
6		EIRE			282			65			23.05			259663.46			961.58			522.47	
35		USA			5			2			Invalid Numb...			1730.92			716.08			491.6	
29		United Arab Emirates			2			0			Invalid Numb...			1864.78			932.39			490.5	
20		Denmark			18			2			Invalid Numb...			18042.14			1011.74			455.17	
21		Cyprus			14			3			Invalid Numb...			12931.29			946.76			454.21	
28		Lebanon			1			0			Invalid Numb...			1693.88			1693.88			386	
Rows: 38																					↗ Expand

Conclusions

Key Insights

Business Growth and Profitability

The business demonstrates a clear upward revenue trend, particularly in peak seasonal periods. However, returns are an existing obstacle pressing profitability in specific markets and large-volume orders. The small number of large-volume transactions takes up a disproportionately high share of returns, amplifying financial risk due to the uneven scale effect. The growth is increasingly offset by rising returns, which becomes the key factor limiting margin expansion.

Localised Nature of Returns

Returns are not evenly distributed across the business but are concentrated in specific markets and products, indicating that the problem is operationally localised, not systemic. This creates an opportunity for targeted interventions instead of general, one-size-fits-all solutions.

Market-level Segmentation: Value vs Risk

There are apparent structural differences existing across markets. The UK remains a stable revenue foundation, supported by its high transaction volume and relatively consistent performance, which should be maintained and optimised as the core market. At the same time, international markets present a more complex picture. While they offer strong revenue potential due to higher order values and bulk purchasing behaviour contributing to higher revenue potential, they also carry considerably higher risks, driven by elevated return rates and large order sizes. This means that high-value bulk purchasing can act as a double-edged sword, providing business opportunities for revenue growth and financial threats when returns occur. Across the international markets, the Netherlands emerges as a high-value, low-risk market, combining strong revenue contribution with minimal return rates. Meanwhile, markets such as Switzerland contrastingly exhibit high return rates alongside large order sizes, indicating high-risk, high-volatility profiles.

Churn Risk Analysis

Although returns do not directly translate into churn, the return behaviour could be used as a proxy indicator measuring customer dissatisfaction and potential churn risk. High-return or high-risk markets could demonstrate potential churn risk due to expectation or operational mismatches. In the meantime, the UK and the Netherlands indicate a much more stable picture in purchasing and return patterns, suggesting stronger customer retention and lower churn likelihood, given the fact that the UK remains the core market.

Strategic Recommendations

Market-focused Growth

For more sustainable growth, balanced, operationally focused and differentiated market strategies should be adopted. The business should continue to prioritise scaling high-performing regions (high-value, low-return), like the Netherlands, while applying more targeted approaches to high-risk markets (large basket sizes and rising return rates such as Switzerland). This includes closely monitoring bulk orders, identifying and implementing strict quality control for high-return products, and applying tailored logistics management moves to improve customer experience as well as reduce damages, delays, or mismatches that could drive returns to manage volatility. For the core, stable market (the UK), the business should continue to maintain and optimise current strategies to enhance this stability.

Return Reduction

Reducing return rates should be the first business priority, as they fundamentally impact both profitability and long-term customer retention. This requires refining product quality, improving fulfilment accuracy and clarity of product information, developing comprehensive solutions for problems customers might encounter during actual product delivery and post-purchase processes to enhance customer experience, and strengthening cross-border logistics procedures, as most of the returns are international. By addressing these issues, the business can simultaneously protect the margins, grow into middle- to high-value segments, and reduce potential churn risk as well as strengthen retention.

Bulk Transaction Control

Since a small number of large-volume transactions account for disproportionately to overall returns, the business should implement stricter monitoring and control mechanisms for bulk orders. This entails early detection of high-risk transaction patterns, validation checks, and the implementation of safeguards to reduce possible financial exposure from large-scale returns.